Varendra University

Dept. of Computer Science & Engineering (CSE)

The String Instructions (Chapter-11)

Course Title: **Microprocessor and Assembly Language**
Course No: **CSE 3205**

Prepared By:
Md. Adnan Sami
Lecturer, Dept. of CSE, VU

Outline

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General Form of the String Instructions

Instruction	*Destination*	*Source*	*Byte form*	*Word form*
Move string	ES:DI	DS:SI	MOVSB	MOVSW
Compare string	ES:DI	DS:SI	CMPSB	CMPSW
Store string	ES:DI	AL or AX	STOSB	STOSW
Load string	AL or AX	DS:SI	LODSB	LODSW
Scan string	ES:DI	AL or AX	SCASB	SCASW

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General Form of the String Instructions

Instruction		*Example*
MOVS	destination_string, source_string	MOVSB
CMPS	destination_string, source_string	CMPSB
STOS	destination_string	STOS STRING2
LODS	source_string	LODS STRING1
SCAS	destination_string	SCAS STRING2

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Moving a String

The MOVSB instruction,

```
MOVSB                    ;move string byte
```

```
MOV AX,@DATA
MOV DS,AX                ;initialize DS
MOV ES,AX                ;and ES
LEA SI,STRING1           ;SI points to source string
LEA DI,STRING2           ;DI points to destination string
CLD                      ;clear DF
MOVSB                    ;move first byte
MOVSB                    ;and second byte
```

MOVSB

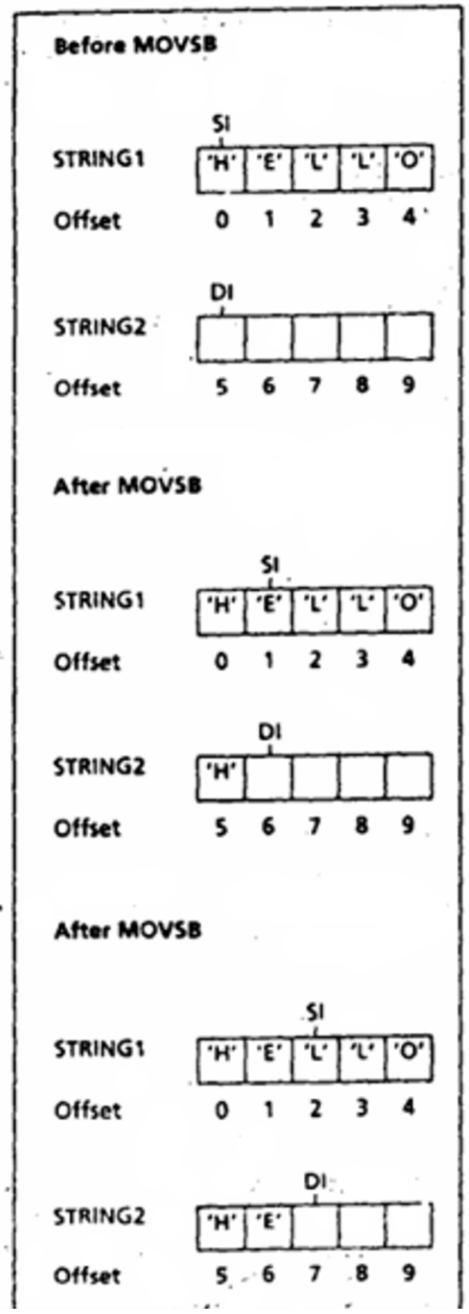


Figure 11.1

MOVSW

There is a word form of MOVSB. It is

```
MOVSW      ;move string word
```

MOVSW moves a word from the source string to the destination string.

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Example 11.2 (*)

For the following array,
ARR DW 10, 20, 40, 50, 60, ?

Write instructions to insert 30 between 20 and 40. (Assume DS and ES have been initialized to the data segment.)

Solution:

```
STD                          ;right to left processing
LEA SI,ARR+8h                ;SI pts to 60
LEA DI,ARR+Ah                ;DI pts to ?
MOV CX,3                     ;3 elts to move
REP MOVSW                    ;move 40,50,60
MOV WORD PTR [DI],30         ;insert 30
```

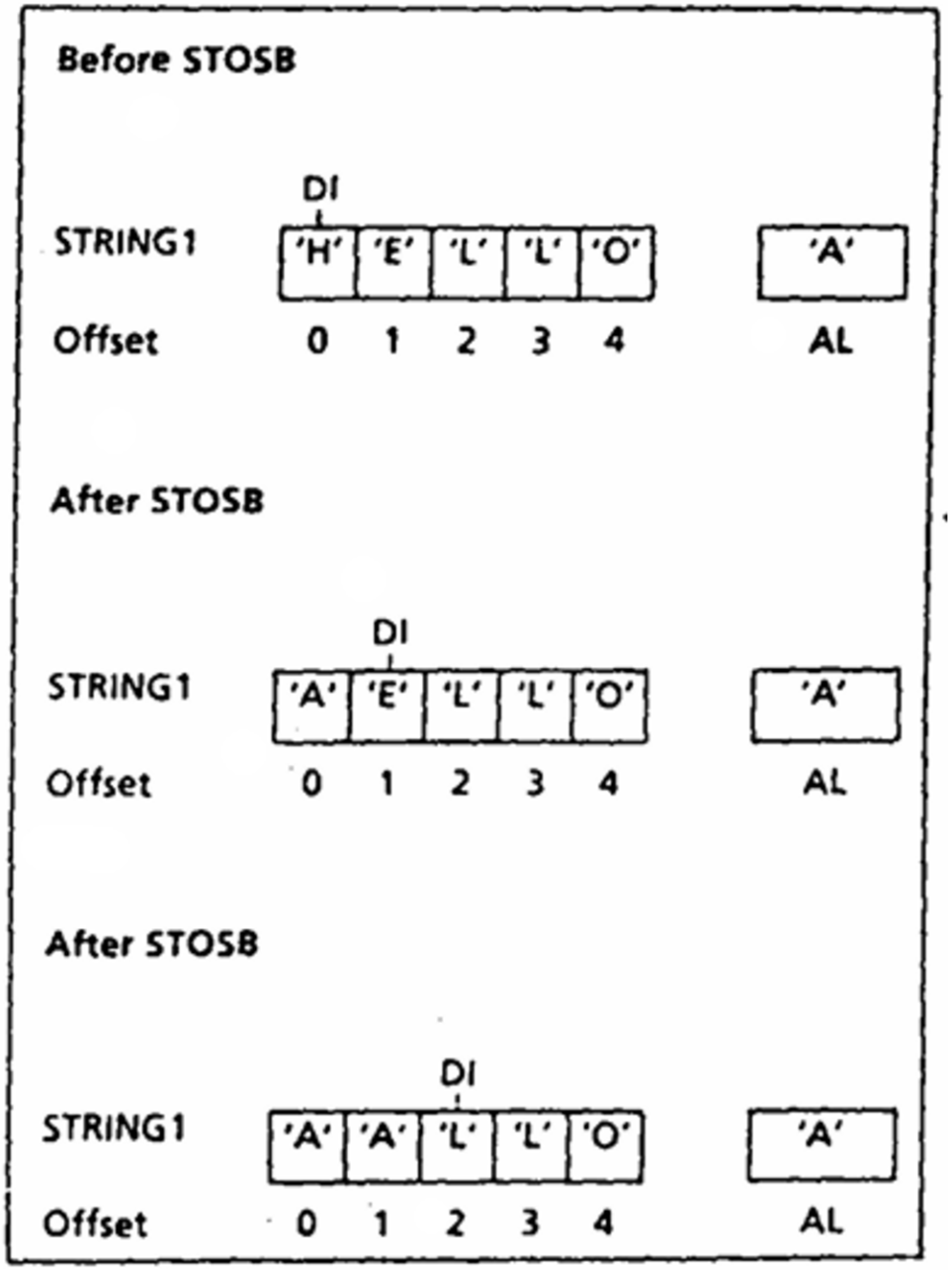
STOSB

```
STOSB           ;store string byte

STOSW           ;store string word
```

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STOSB


Before STOSB
DI
STRING1
'H' 'E' 'L' 'L' 'O'
'A'
Offset
0 1 2 3 4
AL
After STOSB
DI
STRING1
'A' 'E' 'L' 'L' 'O'
'A'
Offset
0 1 2 3 4
AL
After STOSB
DI
STRING1
'A' 'A' 'L' 'L' 'O'
'A'
Offset
0 1 2 3 4
AL

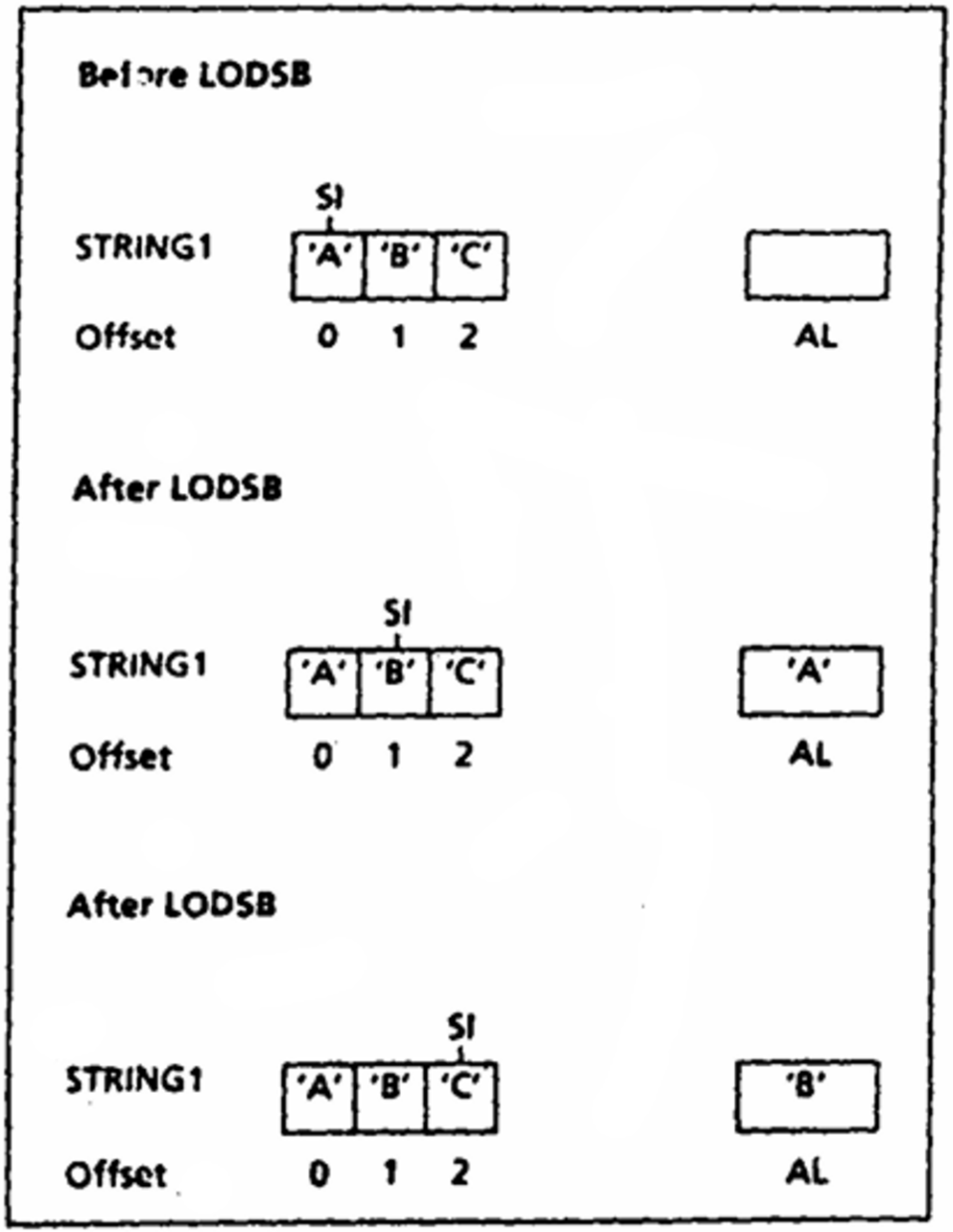
LODSB

```
LODSB                    ;load string byte
```

```
LODSW                    ;load string word
```

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LODSB


Before LODSB
SI
STRING1
'A' 'B' 'C'
Offset
0 1 2
AL
After LODSB
SI
STRING1
'A' 'B' 'C'
Offset
0 1 2
'A'
AL
After LODSB
SI
STRING1
'A' 'B' 'C'
Offset
0 1 2
'B'
AL

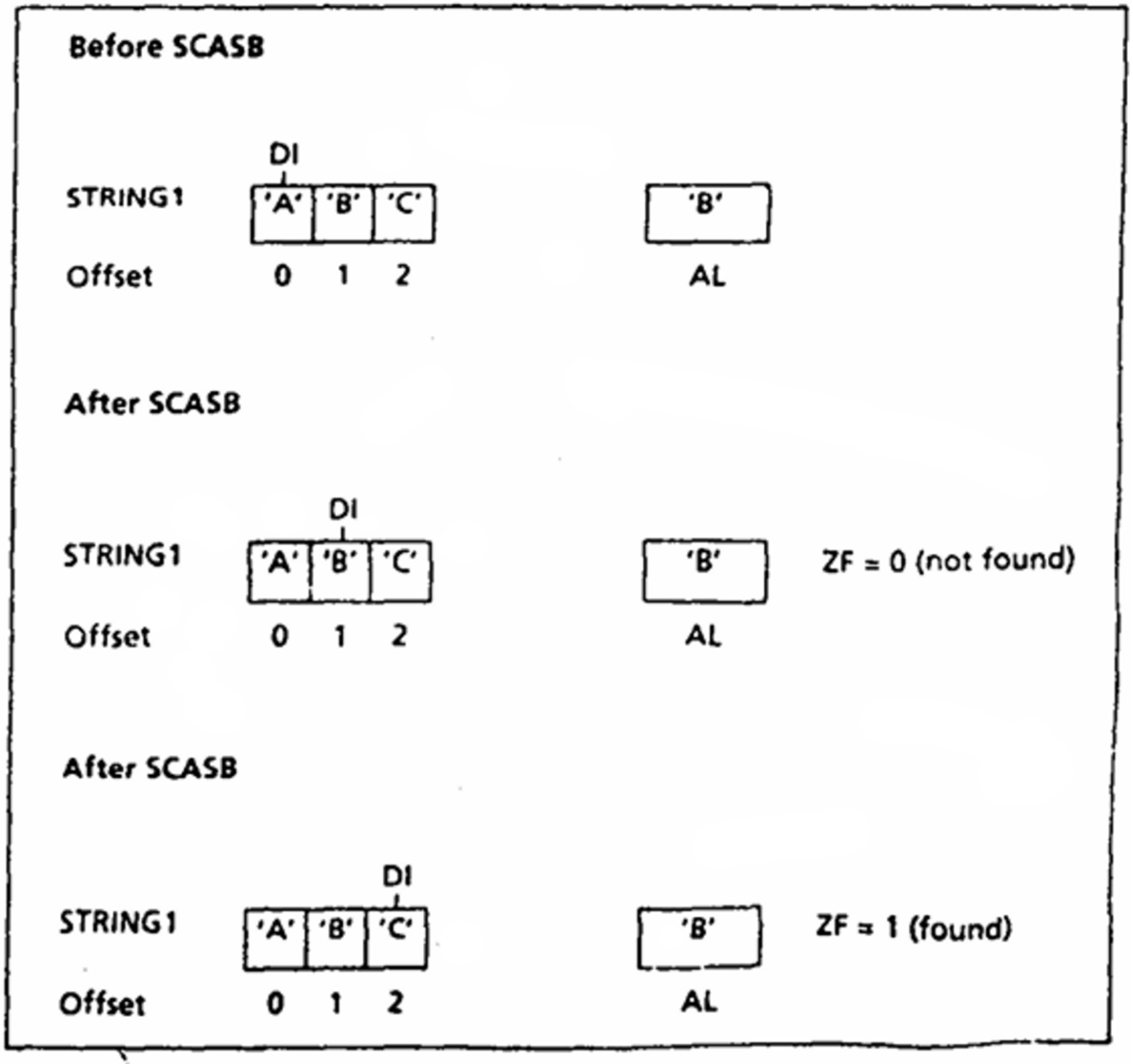
SCASB

```
SCASB     ;scan string byte
```

```
SCASW     ;scan string word
```

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SCASB


Before SCASB
DI
STRING1
'A' 'B' 'C'
Offset
0 1 2
'B'
AL
After SCASB
DI
STRING1
'A' 'B' 'C'
Offset
0 1 2
'B'
AL
ZF = 0 (not found)
After SCASB
DI
STRING1
'A' 'B' 'C'
Offset
0 1 2
'B'
AL
ZF = 1 (found)

CMPSB

```
CMPSB                    ;compare string byte
```

```
CMPSW                    ;compare string word
```

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CMPSB

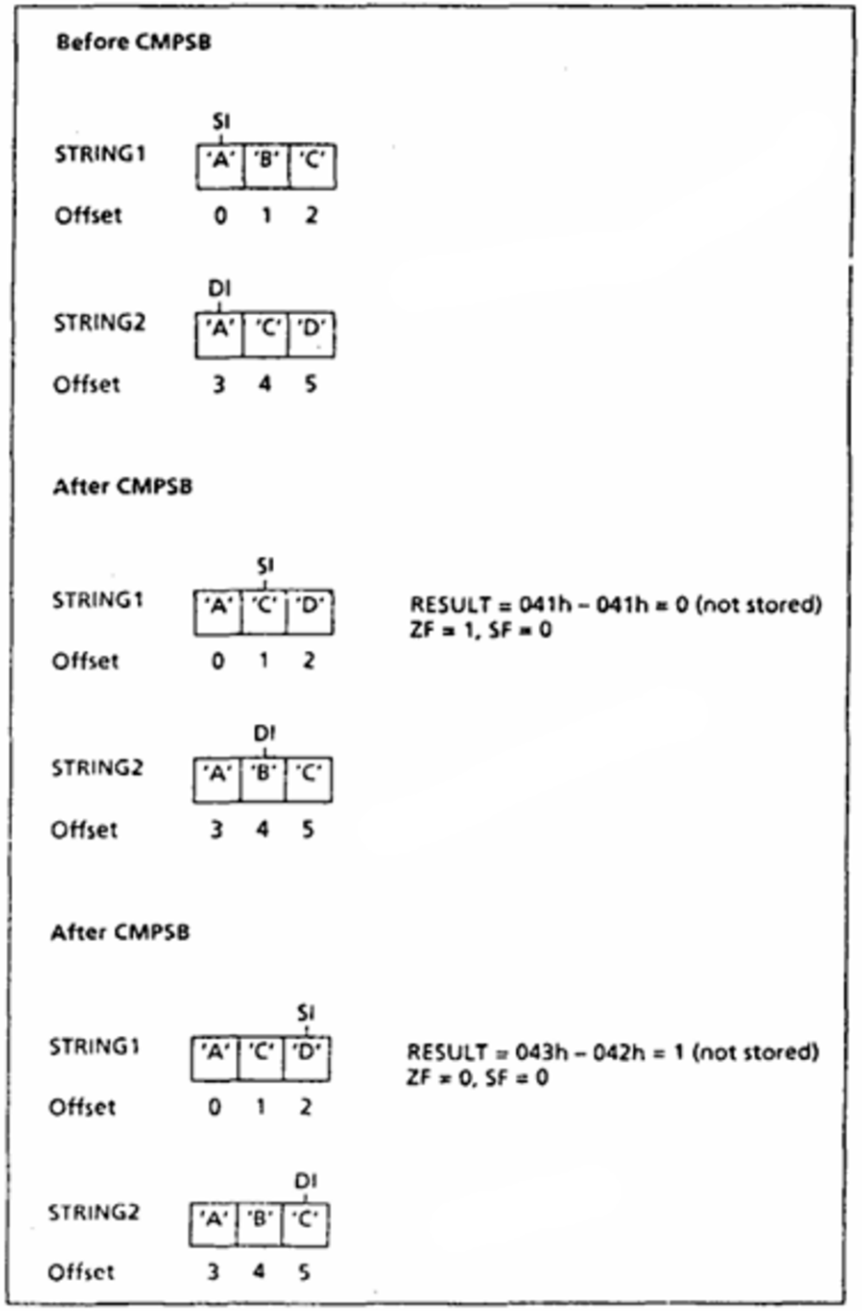


Figure 11.5